

SATHWIK H NAIK

+1(240)-438-0411 | sathwikhnaik@gmail.com | [LINKEDIN](#)

Education

University of Maryland, College Park

2024 - 2026

Master of Science, Data Science (GPA: 3.9)

• **Coursework:** Statistical Inference, Machine Learning, Data Modeling, Data Visualization, Experimental Design

Reva University, India

2019 - 2023

Bachelor of Technology, Computer Science and Engineering (GPA: 3.55)

India

Skills

• Python, R, SQL, Excel, Data Analysis, Data Visualization, Tableau, Power BI, DAX, KPI dashboards, Operational reporting, Git, Salesforce CRM, JIRA, Data Pipelines, Statistical Methods, Monitoring, Logging, CI/CD

Work Experience

Carelon Global Solutions (an Elevance Health subsidiary) | Associate Software Engineer Aug 2023 - Aug 2024

- Gathered business requirements and translated operational questions into KPIs and recurring reporting for stakeholders.
- Built and maintained dashboards and reports (Power BI) to communicate performance metrics and improve visibility into operational health.
- Sourced and validated data for reporting using SQL-based analysis and structured data cleaning, ensuring accuracy and consistency across outputs.
- Designed analytics deliverables (dashboards, recurring reports) that reduced manual effort and enabled faster decision-making for internal teams.
- Documented report logic, KPI definitions, and refresh processes to support self-serve access and reduce ambiguity for stakeholders.
- Collaborated cross-functionally to align reporting and analytics deliverables with team priorities and timelines.
- Improved operational reliability by automating deployment workflows, reducing release cycle time by ~30% and strengthening monitoring discipline.
- Supported controlled, audit-friendly processes through change tracking, documentation, and post-release monitoring aligned with organizational risk practices.

Projects

Conditional Generative Data Augmentation for Image Classification

May 2026

- Developed Conditional GAN (CGAN) and VAE-GAN pipelines to generate synthetic training data and evaluated their impact on MNIST and Fashion-MNIST image classification.
- Conducted architecture, training-strategy, and data-scarcity ablation studies across 12 experimental settings, identifying factors affecting synthetic-data quality and downstream model performance.
- Improved Fashion-MNIST accuracy from 63.6% to 70.7% (+7.1 pp) under extreme data scarcity and showed that synthetic augmentation effectiveness depends on generator quality, dataset complexity, and mixing strategy.

TINYACE: Context Engineering for Small Language Models

Dec 2025

- Co-developed TINYACE, a bounded working-memory framework for small language models to manage context growth under strict token and latency constraints.
- Conducted multi-scale evaluation and ablation studies across TinyLlama (1.1B), Phi-3 Mini (3.8B), and Mistral (7B), identifying a performance “capacity sweet spot” for mid-sized models.
- Built context management strategies (FIFO vs utility scoring) and evaluated trade-offs between accuracy, latency, and safe updates, mirroring how operational rules shape trustworthy automation.
- Defined evaluation metrics beyond exact match (semantic similarity and option-mapped accuracy) to improve interpretability and decision trust.

Advanced Time Series Analysis using Google Cloudpickle

May 2025

- Developed a refreshable analytics dashboard with automated ingestion and time-windowed reporting (up to last 24 hours) for trend monitoring.
- Built operational KPIs (moving averages, trend signals) and visualizations to highlight anomalies and performance changes over time.
- Created a reproducible workflow (Docker + Git) and documented setup/run steps for consistent reporting delivery.

Real-time Sign Language Detection using Convolutional Neural Network

Jun 2023

- Built a real-time gesture recognition tool using TensorFlow + OpenCV.
- Achieved high frame-rate accuracy in American Sign Language (ASL) recognition; user-friendly UI added for accessibility.

Certifications

- Introduction to Data Science in Python: University of Michigan
- Machine Learning: KTH Royal Institute of Technology
- Foundations of Project Management: Google